



GRADE 9
Computational Thinking

National Level Examination (NLE) 2026

SUBJECT CODE

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Total Questions: 50

Total Marks: 50

Time: 75 Minutes

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And Hb Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Abstract & Spatial Thinking (10 Questions); Section B: Pattern Recognition (10 Questions); Section C: Decomposition, Data & Constraints (10 Questions); Section D: Algorithmic Thinking & Logical Flow (10 Questions); Section E: Ai Literacy & Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

SECTION A – ABSTRACT & SPATIAL THINKING

Q1. What is the decimal value of the binary number 110101?

- A. 49
- B. 51
- C. 53
- D. 55

Q2. The point $(-4, 3)$ is rotated 90° clockwise about the origin. What are its new coordinates?

- A. $(-3, -4)$
- B. $(3, 4)$
- C. $(-3, 4)$
- D. $(4, 3)$

Q3. A $4 \times 4 \times 4$ cube is painted on all outer faces and cut into unit cubes. How many unit cubes have exactly two painted faces?

- A. 16
- B. 20
- C. 24
- D. 32

Q4. All objects in set A are in set B, and no object in set B is in set C. Which statement must be true?

- A. Some A are C
- B. No A is C
- C. All C are A
- D. A and C are identical

Q5. Which binary number represents the decimal number 73?

- A. 1000001
- B. 1001001
- C. 1010001
- D. 1011001

Q6. In a group of 80 students, 45 study Python, 30 study AI and 15 study both. How many study neither?

- A. 10
- B. 15
- C. 20
- D. 25

Q7. The binary no.11010 equals to

- A. 56
- B. 57
- C. 58
- D. 60

Q8. What is the binary value of decimal 46?

- A. 101010
- B. 101110
- C. 110010
- D. 111000

Q9. A $5 \times 5 \times 5$ cube is painted on all six faces and cut into unit cubes. How many unit cubes have no painted face?

- A. 8
- B. 27
- C. 36
- D. 64

Q10. The point $(-2, -5)$ is reflected across the x-axis. What are its new coordinates?

- A. $(2, -5)$
- B. $(-2, 5)$
- C. $(5, -2)$
- D. $(-5, 2)$

SECTION B – PATTERN RECOGNITION

Q11. Find the next number: 2, 5, 12, 27, 58, ____.

- A. 116
- B. 119
- C. 121
- D. 123

Q12. Find the next number: 3, 8, 15, 24, 35, ____.

- A. 44
- B. 46
- C. 48
- D. 50

Q13. Find the next number: 1, 4, 13, 40, 121, ____.

- A. 242
- B. 343
- C. 364
- D. 366

Q14. Find the next number: 6, 12, 11, 22, 21, 42, 41, ____.

- A. 80
- B. 81
- C. 82
- D. 84

Q15. A rule maps n to $n(n + 1)$. If $2 \rightarrow 6$, $3 \rightarrow 12$ and $5 \rightarrow 30$, what is $8 \rightarrow$?

- A. 64
- B. 70
- C. 72
- D. 80

Q16. B, C, E, G, K, M, Q correspond to prime-numbered alphabet positions. Which letter comes next?

- A. R
- B. S
- C. T
- D. U

Q17. Find the next number: 5, 11, 23, 47, 95, ____.

- A. 181
- B. 189
- C. 191
- D. 193

SECTION B – PATTERN RECOGNITION (CONTINUED)

Q18. Find the next number: 2, 3, 6, 11, 18, 27, ____.

- A. 36
- B. 38
- C. 40
- D. 42

Q20. Find the next number: 1, 1, 2, 3, 5, 8, 13, ____.

- A. 18
- B. 20
- C. 21
- D. 22

Q19. Find the next hexadecimal value: A, C, F, 13, 18, ____.
(values shown in base 16)

- A. 1C
- B. 1D
- C. 1E
- D. 1F

SECTION C – DECOMPOSITION, NUMBERS & DATA

Q21. A quantity increases by 20% and then decreases by 20%. If it starts at 500, what is its final value?

- A. 480
- B. 490
- C. 500
- D. 520

Q26. A model classifies 500 cases: 360 of 400 Class-A cases and 70 of 100 Class-B cases are correct. What is its overall accuracy?

- A. 82%
- B. 84%
- C. 86%
- D. 88%

Q22. The mean of 14, 18, 21, 27 and x is 22. What is x?

- A. 28
- B. 30
- C. 32
- D. 34

Q27. A fair die is rolled once. What is the probability of getting a prime number?

- A. $\frac{1}{3}$
- B. $\frac{1}{2}$
- C. $\frac{2}{3}$
- D. $\frac{5}{6}$

Q23. A dataset of 1200 records is divided 70% training, 15% validation and 15% testing. How many records are in the test set?

- A. 150
- B. 180
- C. 210
- D. 240

Q28. A dataset has values 4, 5, 5, 6, 10. What is the median?

- A. 5
- B. 6
- C. 6.5
- D. 10

Q24. A 64×64 grayscale image stores 8 bits per pixel. Ignoring metadata, how many bytes are needed?

- A. 2048
- B. 4096
- C. 8192
- D. 32768

Q29. A 256×256 RGB image uses 24 bits per pixel. Ignoring metadata, how many bytes are needed?

- A. 65,536
- B. 131,072
- C. 196,608
- D. 262,144

Q25. Red and blue tokens are in the ratio 3 : 5, with 160 tokens total. If 25% of the blue tokens are changed to red, what is the new ratio red : blue?

- A. 15 : 17
- B. 17 : 15
- C. 7 : 5
- D. 5 : 3

Q30. If 18 of 60 predictions are wrong, what is the model's accuracy?

- A. 60%
- B. 65%
- C. 70%
- D. 75%

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING

Q31. Start with $x = 4$. For $i = 1, 2, 3$, update $x = x \times i + 1$. What is the final value of x ?

- A. 30
- B. 32
- C. 34
- D. 36

Q33. Variables have $a = 7$ and $b = 11$. Execute $\text{temp} = a$; $a = b$; $b = \text{temp}$. What are the final values?

- A. $a=7, b=11$
- B. $a=11, b=7$
- C. $a=11, b=11$
- D. $a=7, b=7$

Q32. Apply this rule three times: if x is even, $x = x/2$; otherwise $x = 3x + 1$. Start with $x = 5$. What is x after three steps?

- A. 2
- B. 4
- C. 8
- D. 16

Q34. A nested loop runs j from 1 to i for each i from 1 to 4. How many times does the inner action run in total?

- A. 8
- B. 9
- C. 10
- D. 16

SECTION D – ALGORITHMIC THINKING & LOGICAL SEQUENCING (CONTINUED)

Q35. A weighted network has edges A-B=2, A-C=6, B-C=1, B-D=5 and C-D=2. What is the minimum cost from A to D?

- A. 4
- B. 5
- C. 6
- D. 7

Q36. An average algorithm sets sum=0 inside the loop before adding each new value. What is the correct fix?

- A. Set sum=0 once before the loop
- B. Set sum=1 inside the loop
- C. Divide before adding
- D. Remove the loop

Q37. Set $s = 0$. For $i = 1$ to 6, if i is odd add i^2 to s ; otherwise add i . What is s ?

- A. 38
- B. 41
- C. 44
- D. 47

Q38. A binary search is used on a sorted list of 64 items. What is the maximum number of halving steps needed to reduce to one item?

- A. 5
- B. 6
- C. 7
- D. 8

Q39. A shortest grid path needs 4 Right moves and 3 Up moves. How many shortest paths are possible?

- A. 20
- B. 30
- C. 35
- D. 42

Q40. A loop should run for $i = 1$ to 10, but the condition is written $i < 10$. How many times does it run?

- A. 8
- B. 9
- C. 10
- D. 11

SECTION E – OLYMPIAD CHALLENGE

Q41. An AI predicts a student's numerical score from previous performance data. Which technique is most suitable?

- A. Classification
- B. Regression
- C. Clustering
- D. Encryption

Q42. A classifier marks 50 items as positive; 40 of those are actually positive. What is its precision?

- A. 70%
- B. 75%
- C. 80%
- D. 90%

Q43. An AI groups documents by similarity without predefined labels. Which method is being used?

- A. Classification
- B. Regression
- C. Clustering
- D. Sorting only

Q44. A medical AI is trained on data that under-represents one population. What should be checked most carefully before deployment?

- A. Font size
- B. Bias and fairness across groups
- C. Screen brightness
- D. File extension

Q45. Which practice best protects privacy when collecting data for an AI system?

- A. Collect every possible detail
- B. Collect only necessary data with informed consent
- C. Share raw data publicly
- D. Keep passwords in the dataset

Q46. A 4-digit code uses 1, 2, 3 and 4 once each. The first digit is twice the second, first + last = 5, third > last, and third < first. What is the code?

- A. 2143
- B. 2413
- C. 4231
- D. 4312

Q47. A model performs very well on training data but poorly on new data. This is most likely:

- A. Clustering
- B. Overfitting
- C. Encryption
- D. Compression

Q48. Which metric measures the fraction of actual positives that a classifier correctly finds?

- A. Precision
- B. Recall
- C. Mean
- D. Variance

Q49. Which AI domain is central to understanding and generating human language?

- A. Computer Vision
- B. Natural Language Processing
- C. Robotics only
- D. File Management

Q50. A generative-AI answer sounds confident but contains false facts. What is the best response?

- A. Accept it because it sounds certain
- B. Verify important claims using reliable sources
- C. Share it immediately
- D. Remove all source citations